Lecture plan · for the lecturer

Designing a data-science module

Data science · University · Lecture 4

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Align outcomes and assessment.
- Design a staged, real project.
- Write a rubric that teaches.

Activity: Align a module

1. Warm-up (5 min): Ask colleagues to sort last year's exam questions into 'calculate' and 'judge'.
2. Teach (10 min): Show an alignment table and the six GAISE recommendations.
3. Map (20 min): Pairs map their own module's outcomes to its assessments and mark the gaps.
4. Draft (20 min): Write four criteria and four levels for a project, then swap and critique.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which outcome has no assessment?
- How will you check reproducibility at scale?
- What feedback will the draft get?

Exit ticket

- Write one outcome with an active verb and the task that assesses it.

Try it in your next lecture
Take your module's learning outcomes, underline each verb, and write next to it the assessment that measures it; any verb with nothing beside it needs a new task.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **When should students first see the project rubric?**

◯ With their final grade ◯ When the project is set ◯ Never, to avoid gaming it

2 **Complete the weighting.**

A module gives 40% to the project and 25% to labs. If the exam gets the rest, it is worth _____ %. If the project's report is half its weight, the report is worth _____ %.

Word bank: 25 · 35 · 20 · 40

3 **Which outcomes are written so they can be assessed? Choose all that apply.**

☐ Appreciate the role of data in society ☐ Critique the causal claim in a published study

☐ Justify the choice of a model for a given dataset ☐ Understand regression

4 **Requiring a reproducible notebook makes copying results from another group harder.**

◯ True ◯ False

5 **Match each piece of feedback to Hattie and Timperley's question it answers.**

"The goal is a defensible causal claim."	• • • •	Where to next?
"Your DAG omits prior grade."	• • • •	Where am I going?
"Add prior grade and rerun the model."	• • • •	How am I going?

Answers · for the lecturer

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1 **When the project is set**

A rubric shared early guides the work and sharpens draft feedback.

2 **A module gives 40% to the project and 25% to labs. If the exam gets the rest, it is worth 35%. If the project's report is half its weight, the report is worth 20%.**

100 − 40 − 25 = 35, and half of 40 is 20.

3 **Justify the choice of a model for a given dataset; Critique the causal claim in a published study**

Justify and critique describe observable performance; understand and appreciate don't say what a student would do to show it.

4 **True**

When every figure must rerun from raw data with the student's own code, pasted results show up quickly.

5 **"The goal is a defensible causal claim." — Where am I going?**
"Your DAG omits prior grade." — How am I going?
"Add prior grade and rerun the model." — Where to next?

Feedback that answers all three gives a student a goal, a diagnosis and an action.

✎ **For you to think about**

Which assessment in your module rewards calculation where the outcome asks for judgement? What would you replace it with?